

FACULTY OF COMPUTING
— STUDENTS —

Better
Habits
Build a
Better
Campus
♡

NOTICE!



Please **do not**
unplug the PCs.

If you want to use a plug point
to charge your devices,
please use the **spare plugs**
given.

Use the spare plugs
for charging



Keep
Our Labs
Running
😊



Lets protect our labs
and use them with care!

Thank you!
♡



Module	Web Application Development
Practical Number	04
Batch	26.1 IFSLs

Task 01 – Library Membership Registration

Create an HTML webpage named **libraryregistration.html** for a university library membership system.

A. Create the Registration Form

Create a form containing the following fields:

- Full Name - Text input
- Student ID -Text input
- Email Address - Email input
- Year of Study - Drop-down list containing:
 - Year 1 ○ Year 2 ○ Year 3 ○ Year 4
- Membership Type – Radio buttons:
 - Standard
 - Premium
- Receive Library Updates - Checkbox
- Agree to Library Rules -Checkbox
- Create Membership - Button

Apply suitable CSS to improve the appearance of the form.

B. JavaScript Validation

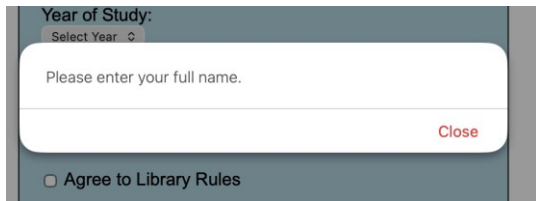
Create a JavaScript function named **validateMembership()** and call the function when the **Create Membership** button is clicked.

The function should validate the following:

1. Full Name must not be empty.
2. Student ID must not be empty.
3. Email Address must not be empty.
4. A Year of Study must be selected.
5. A Membership Type must be selected.
6. The **Agree to Library Rules** checkbox must be selected.

If any validation fails, display an appropriate message using `alert()`.

For example:



Year of Study:
Select Year ▾

Please enter your full name.

Close

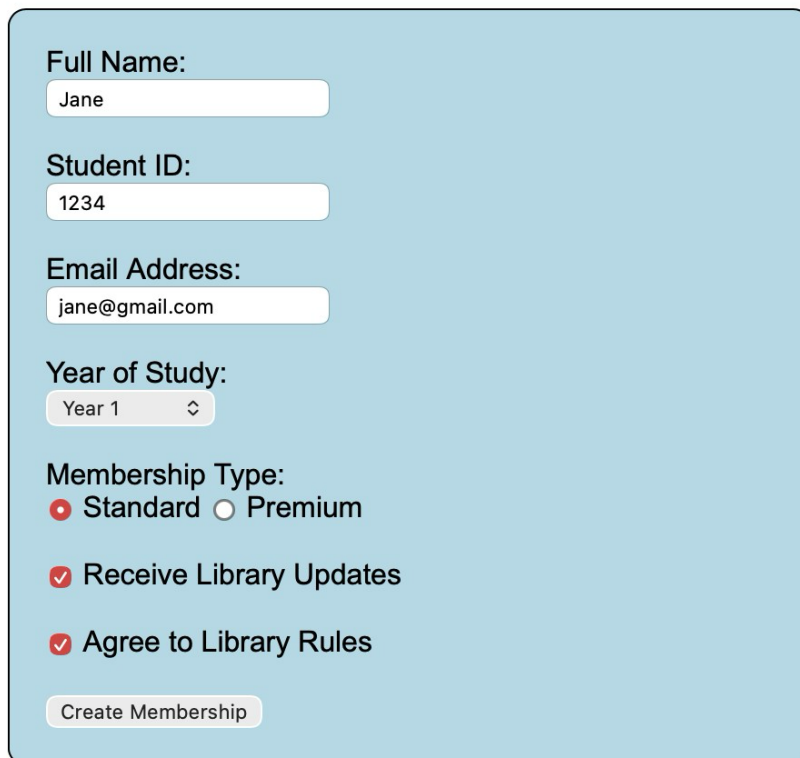
☐ Agree to Library Rules

The **Receive Library Updates** checkbox is optional and should not prevent registration.

C. Display the Result

If all the required information is valid, display the following information **below the form**:

Library Membership Registration



Full Name:

Student ID:

Email Address:

Year of Study:
 ▾

Membership Type:
☒ Standard ☐ Premium

☒ Receive Library Updates

☒ Agree to Library Rules

Membership Registration Successful!

Name: Jane
Student ID: 1234
Year of Study: Year 1
Membership Type: Standard

Task 02 – Student Marks Analyzer

Create an HTML webpage named **marks.html** to analyze the marks obtained by a student.

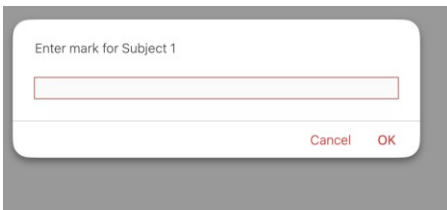
A. Collect the Marks

Create a button named **Enter Marks**.

When the button is clicked, call a JavaScript function named: `analyzeMarks()`

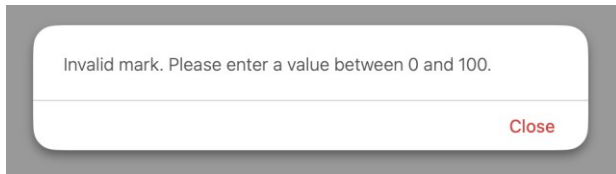
Inside the function, use a **loop** to ask the user to enter marks for **5 subjects**.

For example:

A screenshot of a web browser showing a dialog box. The dialog box has a title bar that says "Enter mark for Subject 1". Inside the dialog, there is a single-line text input field. Below the input field, there are two buttons: "Cancel" and "OK".

Each mark must be between **0 and 100**.

If the user enters a value below 0 or above 100, display:

A screenshot of a web browser showing a dialog box. The dialog box has a title bar that says "Invalid mark. Please enter a value between 0 and 100.". Inside the dialog, there is a single-line text input field. Below the input field, there is a button labeled "Close".

The user should then be asked to enter the mark for that subject again.

B. Calculate the Results

Using the five marks entered by the user, calculate:

- Total marks
- Average mark
- Highest mark
- Lowest mark

A **loop must be used** when processing the marks.

C. Determine the Grade

Use the calculated average to assign the student's grade according to the following criteria:

Average Mark	Grade
75 – 100	A
65 – 74	B
55 – 64	C
40 – 54	S
Below 40	F

Use if, else if, and else statements to determine the grade.

D. Display the Result

Display the final results on the webpage.

Example:

Student Marks Analyzer

Enter Marks

Student Results

Total Marks: 291
Average Mark: 58.2
Highest Mark: 89
Lowest Mark: 34
Grade: C

Task 03 – Simple Online Food Ordering System

Create an HTML webpage named **order.html** for a simple online food ordering system.

A. Create the Order Form

Create a form containing the following customer details:

- **Customer Name** – Text input • **Phone Number** – Text input

Provide the following food items using **checkboxes**:

Food Item Price

Burger \$5

Pizza \$8

Sandwich \$4

Soft Drink \$2

Add a button named **Place Order**.

When the button is clicked, call a JavaScript function named:

placeOrder()

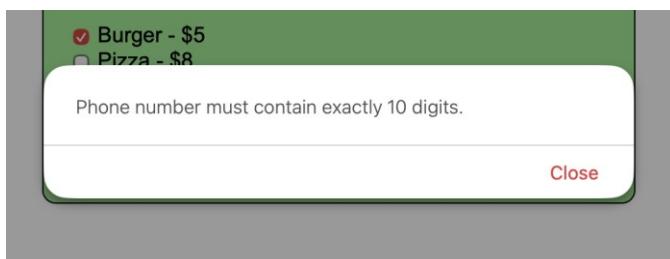
B. Validate the Order

Inside the placeOrder() function, validate the following:

1. Customer Name must not be empty.
2. Phone Number must not be empty.
3. Phone Number must contain exactly **10 digits**.
4. At least **one food item** must be selected.

Display an appropriate error message using alert() if any validation fails.

For example:



C. Calculate the Total Price

Use a **loop** to check the selected food items.

For each selected item, add its price to the total.

For example, if the customer selects:

Burger \$5

Pizza \$8 the

total should be:

Total Price: \$13

D. Display the Order Summary

If all validation is successful, display the order summary **below the form**.

Apply suitable CSS to make the order form and result section clear and readable.

Online Food Ordering System

Customer Name:

Phone Number:

Select Food Items

☒ Burger - \$5
☒ Pizza - \$8
☐ Sandwich - \$4
☐ Soft Drink - \$2

Order Successful!

Customer: Bob
Phone: 0987689765

Selected Items:
Burger
Pizza

Total Price: \$13